PRACTICAL 4
File and Directory Management

Aim

To create, delete, copy and move files and directories.

Create directory

mkdir tyit_lab

cd tyit_lab

Create files

touch file1.txt file2.txt file3.txt

List files

ls -l

Copy file

cp file1.txt file1_backup.txt

Move/rename

mv file2.txt renamed.txt

Create directory

mkdir documents

Copy directory

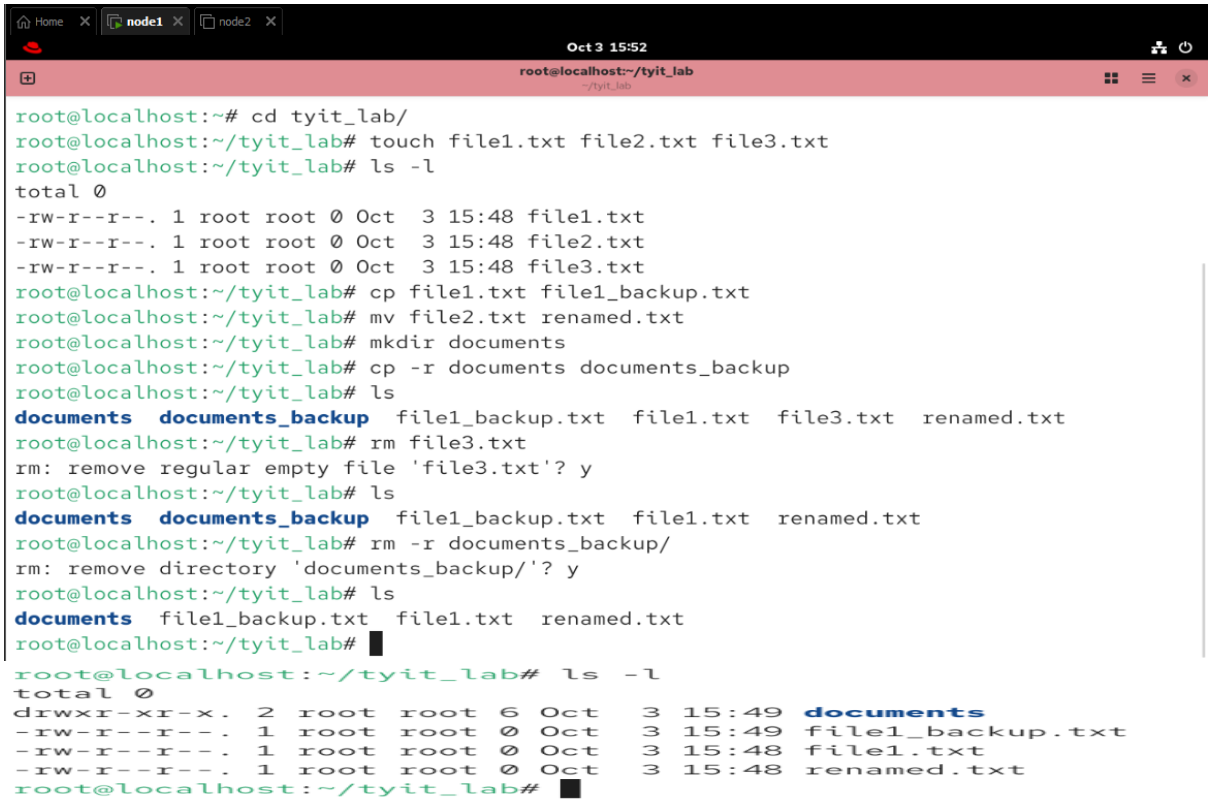
cp -r documents documents_backup

Remove file

rm file3.txt

Remove directory

rm -r documents_backup



PRACTICAL 5

Links, Archives and Compression

Aim

To create hard/soft links and manage compressed and archived files.

A. Hard Link

Create a file:

echo "TYIT Linux" > original.txt

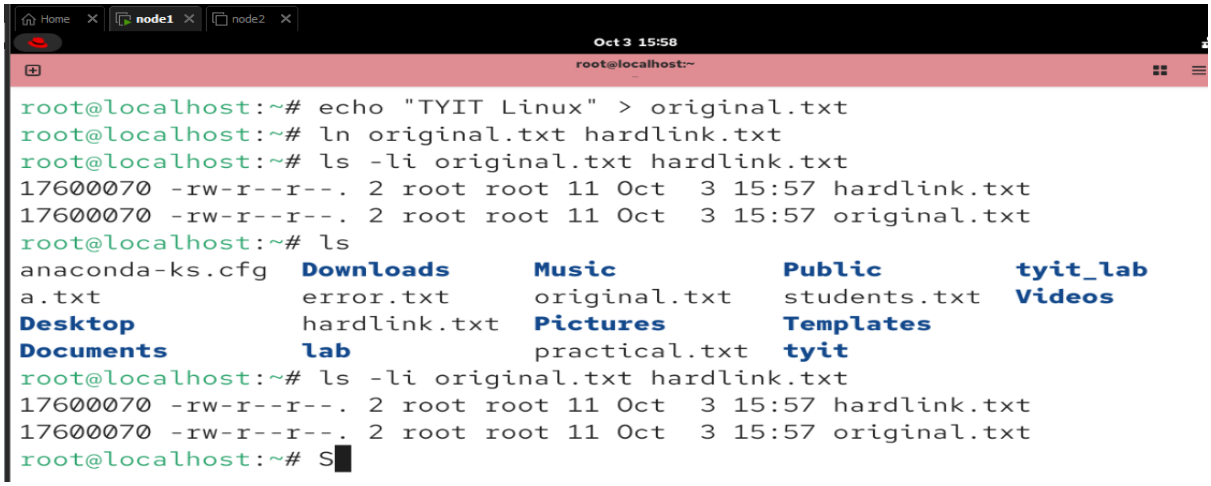
Create hard link:

ln original.txt hardlink.txt

Check inode:

ls -li original.txt hardlink.txt

Both files should have the same inode number.



B. Soft Link

ln -s original.txt softlink.txt

Check:

ls -l softlink.txt

Expected:

softlink.txt -> original.txt

```
root@localhost:~# ln -s original.txt softlink.txt
root@localhost:~# ls -l softlink.txt
lrwxrwxrwx. 1 root root 12 Oct  3 15:59 softlink.txt -> original.txt
root@localhost:~#
```

C. tar

Create archive:

tar -cvf tyit.tar original.txt hardlink.txt

List archive:

tar -tvf tyit.tar

Extract:

mkdir extracted

tar -xvf tyit.tar -C extracted

D. gzip

gzip original.txt

Check:

ls

Uncompress:

gunzip original.txt.gz

E. bzip2

bzip2 original.txt

Uncompress:

bunzip2 original.txt.bz2

F. star

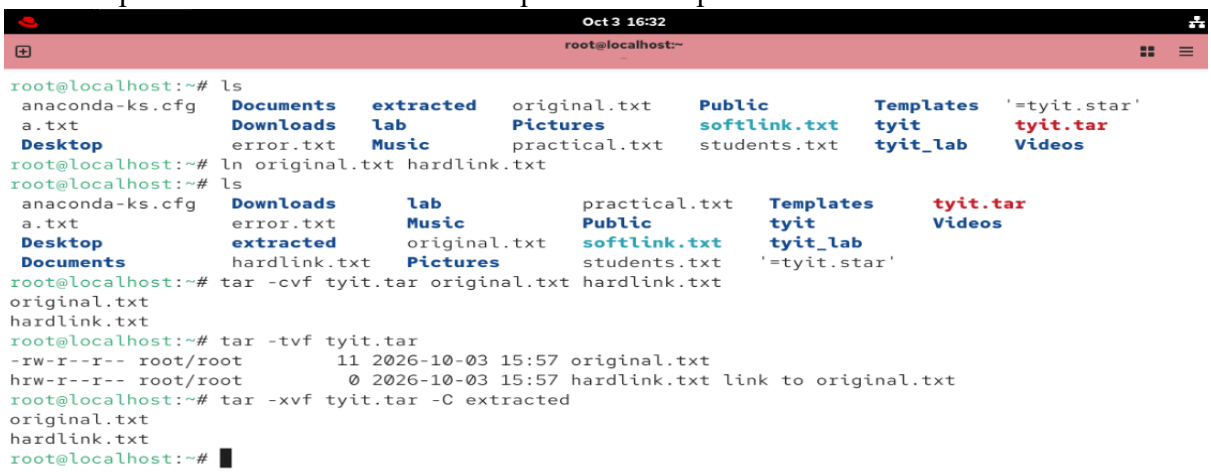
If star is installed:

star -c -f=tyit.star original.txt

List:

star -t -f=tyit.star

If star is not installed in your RHEL installation, install the package through the configured RHEL repositories or demonstrate the equivalent tar operation.



```
root@localhost:~# tar -c -f=tyit.star original.txt
root@localhost:~# star -t -f=tyit.star
bash: star: command not found...
Similar command is: 'tar'
root@localhost:~# tar -t -f=tyit.star
original.txt
root@localhost:~#
```

```
root@localhost:~# gzip original.txt
root@localhost:~# ls
anaconda-ks.cfg  Downloads  Music             Public        tyit
a.txt            error.txt  original.txt.gz   softlink.txt  tyit_lab
Desktop          extracted  Pictures          students.txt  tyit.tar
Documents        lab        practical.txt     Templates     Videos
root@localhost:~# gunzip original.txt.gz
root@localhost:~# ls
anaconda-ks.cfg  Downloads  Music             Public        tyit
a.txt            error.txt  original.txt      softlink.txt  tyit_lab
Desktop          extracted  Pictures          students.txt  tyit.tar
Documents        lab        practical.txt     Templates     Videos
root@localhost:~#
```